Energy Balance with N Atmospheric Layers

Name: _____ Roll number: _____

Constants. Solar constant S = 1361 W m^{-2}; Stefan–Boltzmann constant $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}; planetary albedo $\alpha = 0.30$.

Assumptions. The atmosphere is represented as a stack of N horizontal layers. Each layer is transparent to incoming solar radiation and completely opaque to outgoing infrared radiation. Each layer emits as a blackbody, equally upward and downward, and the planet is in radiative equilibrium.

Question 1

With no atmosphere present, the planet absorbs solar radiation over its cross-sectional area and emits over its full surface area. This gives the emission temperature

$$T_e = [\, S(1-\alpha) / (4\sigma) \,]^{1/4}$$

Evaluate T_e.

T_e = _____ K

Question 2

For a single opaque layer, the surface temperature is $T_s = 2^{1/4}\, T_e$.

T_s = _____ K

Question 3

You now have two cases: $N = 0$ (Question 1) and $N = 1$ (Question 2). Before performing any further calculation, propose a general expression for N layers, write it down, and use it to fill in the multipliers below.

General expression: T_s = _____

T_s = _____ × T_e ($N = 1$)

T_s = _____ × T_e ($N = 2$)

T_s = _____ × T_e ($N = 3$)

Question 4

Justify your proposal without solving the system of layer-by-layer energy balances.

In equilibrium the planet must radiate $S(1 - \alpha)/4$ to space. Consider the net upward flux crossing any horizontal level within the stack. State what that flux must equal and use it to obtain the general expression in a single line of working.